EXPERIMENT - 10

AIM :

To write a Python program to implement the A* search algorithm to find the least-cost path from a starting node to a goal node using path cost and heuristic estimation..

ALGORITHM:

1. Start.
2. Initialize the starting node and goal node.
3. Set the starting node's path cost to zero.
4. Calculate the evaluation function using \(f(n)=g(n)+h(n)\), where \(g(n)\) is the path cost and \(h(n)\) is the heuristic estimate.
5. Select the node with the lowest evaluation value.
6. If it is the goal node, display the solution path and total cost.
7. Otherwise, expand the node and calculate the costs of its neighboring nodes.
8. Update the costs and parent information of the neighboring nodes.
9. Repeat until the goal is reached or no nodes remain to explore.
10. Stop.

PROGRAM :

untitled (3.14.3)

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```
import heapq

def astar(graph, heuristic, start, goal):
    queue = [(heuristic[start], 0, start, [start])]
    best_cost = {start: 0}

    while queue:
        f, cost, node, path = heapq.heappop(queue)

        if node == goal:
            return path, cost

        if cost > best_cost.get(node, float("inf")):
            continue

        for neighbor, weight in graph[node]:
            new_cost = cost + weight

            if new_cost < best_cost.get(
                    neighbor, float("inf")):
                best_cost[neighbor] = new_cost
                new_f = new_cost + heuristic[neighbor]

                heapq.heappush(
                    queue,
                    (new_f, new_cost, neighbor,
                     path + [neighbor])
                )

    return None, float("inf")

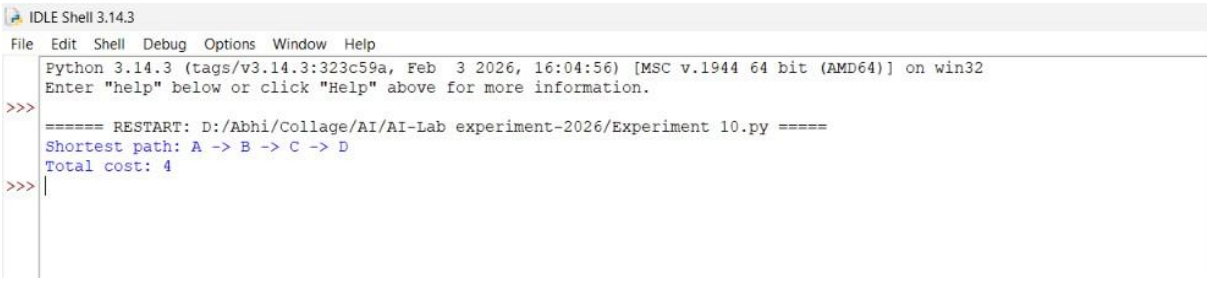
graph = {
    'A': [('B', 1), ('C', 4)],
    'B': [('A', 1), ('C', 2), ('D', 5)],
    'C': [('A', 4), ('B', 2), ('D', 1)],
    'D': [('B', 5), ('C', 1)]
}

heuristic = {
    'A': 4,
    'B': 3,
    'C': 1,
    'D': 0
}

path, cost = astar(graph, heuristic, 'A', 'D')

if path:
    print("Shortest path:", " -> ".join(path))
    print("Total cost:", cost)
else:
    print("No path found")
```

OUTPUT:



RESULT:

Thus, the Python program to implement the A* algorithm was executed successfully, and the path from the start node to the goal node was found.